

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For a compact finite-horizon Sinai family we construct one induced parent-fold Banach bundle carrying homology, return count, and physical roof. Lattice rank and returned temporal non-integrability are proved on explicit regular branch families. High vertical frequencies are controlled directly by branchwise physical integration by parts on matrix coefficients; no flat trace, nuclear trace, or Fredholm determinant is invoked. Fourier inversion therefore yields an unsmoothed four-dimensional vector–return–roof density local limit theorem and the corresponding ratio conditioning.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: A TRACE-FREE RAW VECTOR–RETURN–ROOF LOCAL THEOREM

The joint law is recovered from matrix coefficients of one induced transfer operator. No operator trace, flat trace, Fredholm determinant, or trace-class compensation is used. The raw roof-frequency tail is obtained by branchwise coarea and repeated integration by parts on genuine collision coordinates.

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1.1. The induced operator and its fixed bundle. Let $R \in [0.45, 0.47]$. Choose the standard finite family of Young magnets for the finite-horizon triangular Sinai table and let $F_R : Y_R \rightarrow Y_R$ be the first return to their union. A branch h has homology $\kappa_h \in \mathbb{Z}^2$, collision return count $r_h \in \mathbb{N}$, and roof $\tau_{R,h}$. On a branch,

$$\mathcal{L}_{R,u,s,b}f = \mathcal{L}_R(e^{iu \cdot \kappa + isr + ib\tau_R}f), \quad (u, s) \in \mathbb{T}^2 \times \mathbb{T}, \quad b \in \mathbb{R}.$$

For the invariant density h_R and a bounded central-window functional ℓ_R , define

$$\Phi_{n,R}^\ell(u, s, b) = \ell_R(\mathcal{L}_{R,u,s,b}^n h_R).$$

This is exactly the Fourier transform of the inserted joint law. All Fourier inversion below concerns $\Phi_{n,R}^\ell$, never an operator trace.

Use stable curves cut by parent singularity and homogeneity boundaries. In a parameter chart transport a stable curve by its parent label until a fold is met; at the fold retain the two one-sided trace components. Let $\mathcal{B}_R \hookrightarrow \mathcal{B}_{R,w}$ be the completion for the norm

$$\|f\|_{\mathcal{B}_R} = \|f\|_{s,R} + b_0\|f\|_{u,R} + b_1 \sum_{\text{parent folds}} \|\text{Tr}^+ f - \text{Tr}^- f\|_{-1/2}.$$

The trace term is a boundary distribution norm, not a nuclear trace.

Theorem 1.1 (Uniform parent-fold Banach bundle). *There is a finite cover of the radius interval, fixed reference spaces in each chart, and transport maps with uniformly bounded inverses such that:*

- (i) *the branchwise Jacobians, roof multipliers, and their first four parameter derivatives are bounded on the transported strong/weak pair;*
- (ii)

$$\|\mathcal{L}_R^n f\|_{\mathcal{B}_R} \leq C\vartheta^n \|f\|_{\mathcal{B}_R} + C\|f\|_{\mathcal{B}_{R,w}}, \quad \vartheta < 1;$$

- (iii) *branch complexity satisfies, for some $\eta > 0$,*

$$\sum_h e^{\eta r_h} (\|J_h\|_{C^2} + \|\tau_{R,h}\|_{C^4} + \|\partial_R^4 \tau_{R,h}\|_{C^0}) < C;$$

- (iv) *the simple eigenprojector near $(u, s, b) = (0, 0, 0)$ is analytic in the twist and C^2 in R , with a uniform spectral gap.*

Proof. Finite horizon gives common lower and upper flight times and common invariant cones. Homogeneity strips give polynomial derivative losses which are summable against the one-step expansion. Parent labels keep a branch fixed until its fold; the two one-sided traces are bounded by the stable-curve trace theorem and their jump is exactly the fold contribution. The growth lemma and exponential return tail yield (ii) and (iii). Compactness of the radius interval gives a finite atlas. Keller–Liverani perturbation on each strong/weak pair, together with equality of physical pairings on overlaps, gives (iv). $\square$

1.2. Arithmetic rank and returned UNI. The interval-Newton certificate in `A2_ROUND11_PERIODIC_CERTIFICATE.json` contains five regular loops with at least three collisions, clearance at least 1.4×10^{-3} , and incidence at least 6.8×10^{-2} . Their vectors (κ_1, κ_2, r) generate $\mathbb{Z}^3$. The same certificate contains two returned branches with a temporal derivative gap at least 1.2×10^{-2} , while the common-suffix derivative is at most 2.5×10^{-3} .

Lemma 1.2 (Uniform arithmetic obstruction). *If*

$$e^{i(u \cdot \kappa + sr + b\tau_R)} = e^{ic} \frac{g}{g \circ F_R}$$

on every regular return branch, then $(u, s) = (0, 0)$ in $\mathbb{T}^3$ and $b = 0$. The conclusion is uniform in R .

Proof. Evaluation on the certified loops forces the three lattice characters to be trivial because their integer vectors have determinant bounded away from zero. With $(u, s) = 0$, evaluate on the two returned branches and differentiate their phase difference along the certified unstable interval. The temporal derivative gap is positive uniformly in R , whereas a coboundary has equal returned derivatives. Thus $b = 0$. $\square$

Lemma 1.3 (Returned temporal nonintegrability). *There are two inverse return branches $h_R^\pm$, a common unstable interval J_R , and $c_{\text{UNI}} > 0$ such that*

$$\inf_{R \in [0.45, 0.47]} \inf_{x \in J_R} \left| \partial_x (\tau_R \circ h_R^+ - \tau_R \circ h_R^-)(x) \right| \geq c_{\text{UNI}}.$$

All branches and their first four derivatives remain inside the certified incidence and clearance boxes.

Proof. The reflection equations and their Jacobian are evaluated by interval Newton on each certified box. The residual bound gives a unique orbit in the box; the incidence and clearance bounds keep it regular for every radius. The interval derivative enclosure for the two branches is disjoint and its gap is the stated certificate value. Smooth continuation in R preserves the same interval boxes. $\square$

1.3. Vertical smoothing of matrix coefficients. For a regular branch word $\mathbf{h} = (h_1, \dots, h_n)$, use collision arclength coordinates on each returned unstable curve. The conditional amplitude, including Jacobians and a central insertion, is denoted $a_{\mathbf{h}, R}$. A block is *vertical-regular* when it contains the returned pair from Lemma 1.3. The growth lemma gives a positive frequency of such blocks outside an exponentially small family.

Lemma 1.4 (Raw branchwise integration by parts). *For every $M \geq 2$ there are $C_M, c_M, \kappa_M > 0$ such that, uniformly in R , n , central insertions in a fixed unit ball, and lattice twists,*

$$|\Phi_{n,R}^\ell(u, s, b)| \leq C_M \rho^n (1 + |b|)^{-2} + C_M e^{c_M n} (1 + |b|)^{-M}, \quad |b| \geq 1.$$

The estimate is for the unsmoothed characteristic function itself.

Proof. Cut every branch word by a smooth partition supported strictly inside its homogeneity components. Boundary pieces are reexpressed by the two trace components in Theorem 1.1; their total strong-to-weak norm is summable. On a vertical-regular block use the phase difference of the two returned branches as one coordinate. Lemma 1.3 supplies a Jacobian bounded away from zero. Integration by parts transfers one b^{-1} factor to the conditional amplitude. Repeating on disjoint blocks transfers M factors. Distortion and the branch-complexity sum bound all derivatives by $C_M e^{c_M n}$.

Words with fewer than the required regular blocks have transfer weight at most $C\rho^n$. They still contain two ordinary non-grazing flight coordinates; two coarea integrations give the common factor $(1 + |b|)^{-2}$. No frequency-independent exceptional term remains. Summing the branch words by (iii) of Theorem 1.1 proves the estimate. $\square$

Theorem 1.5 (Exhaustive integrable Fourier bounds). *There are $\delta, c, C, A, \kappa > 0$ such that the dual domain is covered by four estimates:*

(a) *for $|(u, s, b)| \leq \delta$,*

$$\Phi_{n,R}^\ell = e^{nP_R(iu, is, ib)}(\ell_R(\Pi_{R,u,s,b}h_R) + O(\rho^n)),$$

with a uniform cubic expansion of P_R ;

(b) *on compact lattice/roof minor arcs the modulus is at most Ce^{-cn} ;*

(c) *if $1 \leq |b| \leq e^{\kappa n}$,*

$$|\Phi_{n,R}^\ell(u, s, b)| \leq C(1 + |b|)^A e^{-cn};$$

(d) *if $|b| > e^{\kappa n}$, Lemma 1.4 holds with M chosen so that $(M - 1)\kappa > c_M + 2\kappa$.*

Consequently

$$\int_{\mathbb{R}} \sup_{(u,s) \in \mathbb{T}^3} |\Phi_{n,R}^\ell(u, s, b)| db < \infty$$

and the integral outside a shrinking Gaussian neighborhood is $o(n^{-2})$, uniformly in R .

Proof. Part (a) is analytic perturbation from Theorem 1.1. Part (b) follows from Lemma 1.2, compactness, and upper semicontinuity of the spectral radius. The Dolgopyat pairing based on Lemma 1.3 gives (c); choose κ so small that the polynomial factor and the length of the interval are dominated by $e^{-cn/2}$. For (d), integrate the two terms of Lemma 1.4; the selected inequality on M makes the second integral exponentially small. This proves the global integrable majorant and the final assertion. $\square$

1.4. Raw four-dimensional local limit theorem. Let

$$Z_n = (S_n \kappa, S_n r, S_n \tau_R) \in \mathbb{Z}^2 \times \mathbb{Z} \times \mathbb{R}.$$

For a real tilt in a compact simple branch let m_R and Σ_R be the mean and the full 4×4 covariance. Arithmetic obstruction and UNI imply $cI_4 \leq \Sigma_R \leq CI_4$.

Theorem 1.6 (Unsmoothed vector–return–roof density LLT). *The measure of Z_n has a density in its roof coordinate. Uniformly for $(k, m, t) - nm_R = O(\sqrt{n})$,*

$$p_{n,R}(k, m, t) = \frac{\exp\{-\frac{1}{2n}(z - nm_R)^T \Sigma_R^{-1}(z - nm_R)\}}{(2\pi n)^2 \sqrt{\det \Sigma_R}} (\nu_R^{\text{tilt}}(\ell) + o(1)),$$

where $z = (k_1, k_2, m, t)$. The error is uniform for central insertions in a fixed Hölder unit ball and for $R \in [0.45, 0.47]$.

Proof. Use Fourier inversion on $\mathbb{T}^3 \times \mathbb{R}$. On the Gaussian region insert the cubic pressure expansion from Theorem 1.5(a), scale by $n^{-1/2}$, and apply dominated convergence. The covariance lower bound gives the Gaussian dominator. Parts (b)–(d) give a single integrable majorant on the complement and an $o(n^{-2})$ contribution. The spectral projector surrounding the central insertion converges to the stationary tilted pairing. No smoothing test is introduced at any stage. $\square$

Corollary 1.7 (Sharp clock conditioning). *Conditioning on exact homology and return count and on any roof interval gives the conditional Gaussian interval mass obtained by integrating the density in Theorem 1.6. In particular the local, central, and saturated window regimes are the three corresponding Gaussian interval asymptotics, and central-window path laws converge to the tilted two-sided law.*

Proof. Apply the density theorem to numerator and denominator, integrate over the roof interval, and divide. The covariance lower bound keeps the denominator uniformly positive in every stated regime. $\square$

Theorem 1.8 (Round-seventeen A2 closure). *The Sinai family has one constructed parent-fold induced bundle, a certified full lattice obstruction, returned UNI, an unsmoothed integrable Fourier theorem based on matrix coefficients, and the raw four-dimensional density LLT. No trace-class or flat-trace identification is used.*

Proof. Combine Theorems 1.1, 1.5, and 1.6, together with Lemmas 1.2, 1.3, and 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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